## 4 Qualitative Analysis

**S** is a mixture of two solids: **FA 6**, which is soluble in water and **FA 7**, which is insoluble in water. Each of **FA 6** and **FA 7** contains one cation and one anion.

**S** does not contain halide ions.

**(a)** Carry out the following tests. Test and identify any gases evolved.
Carefully record your observations in Table 4.1.

If there are no observable changes, write **no observable change**.

Unless otherwise stated, the volumes given below are approximate and should be estimated rather than measured.

**Table 4.1**

| | test | observations |
|---|---|---|
| 1 | Place all of solid **S**, into a boiling tube. Add 15 $cm^3$ of deionised water and shake for 2 minutes.<br><br>Filter the mixture, collecting the filtrate in a test-tube. Keep the filtrate for tests 2 and 3.<br><br>Wash the residue in the filter paper, with a little deionised water. Discard the washings. Keep the residue for test 4. | |
| 2 | Add 1 cm depth of the filtrate to a test tube.<br><br>Add aqueous ammonia. | |
| 3 | Add 1 cm depth of the filtrate to a boiling tube.<br><br>Add aqueous sodium hydroxide.<br><br>Warm the mixture carefully for 2 minutes.<br><br>Add a small piece of aluminium foil and warm the mixture carefully. | |

| | | |
|---|---|---|
| 4 | Using a spatula, transfer some of the solid residue from the filter paper into a test-tube.<br><br>Add 2 cm depth of **FA 2** ($H_2SO_4$) to the same tube.<br><br>Keep this solution for test 5. | |
| 5 | To the resultant solution from test 4, add an equal volume of deionised water and shake the test-tube. Divide the liquid into two equal portions.<br><br>To the first portion, add aqueous sodium hydroxide.<br><br>To the second portion, add aqueous ammonia. | |

[4]

**(b)** Identify the ions in **FA 6** and **FA 7** respectively and state the evidence by completing Table 4.2.

**Table 4.2**

| | | evidence |
|---|---|---|
| **(i)** | cation in **FA 6** | |
| **(ii)** | cation in **FA 7** | |
| **(iii)** | anion in **FA 7** | |

[3]

**(c)** Using your observation in Test 3, suggest the possible identities of the anion in **FA 6.** Hence describe a test to confirm the identity of the anion. You do not need to carry out this test.

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..... [2]

[Total: 9]